

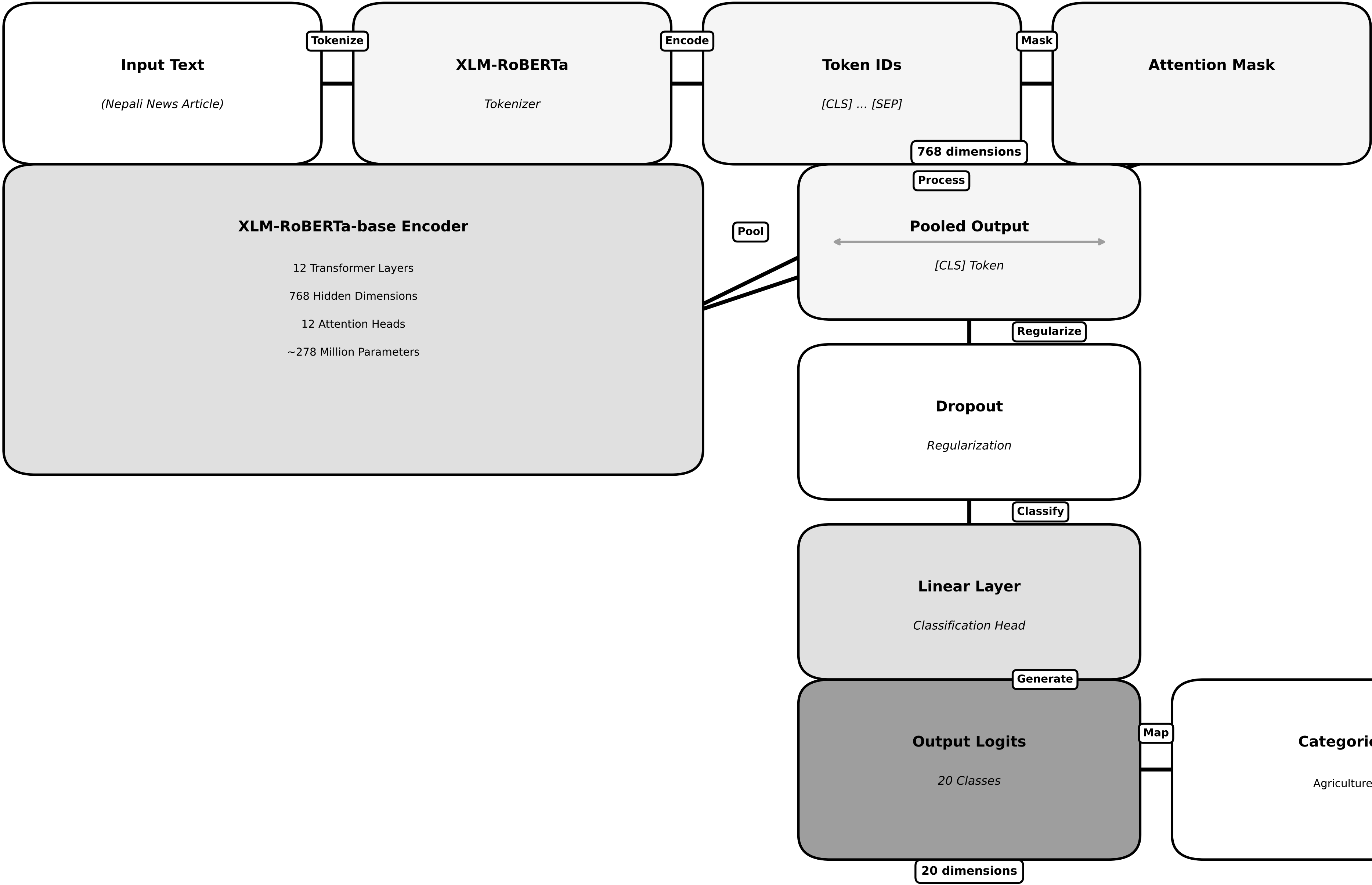
XLM-RoBERTa-based Nepali News Classifier Architecture

Input/Output

Tokenization

XLM-RoBERTa Encoder

Output Layer



Model Configuration

Base Model: XLM-RoBERTa-base

Total Parameters: ~278M

Trainable Parameters: ~278M

Input Format: Nepali text (Devanagari)

Output: 20 news categories

Max Sequence Length: 512 tokens

Training Configuration

Loss Function: CrossEntropyLoss with class weighting

Optimizer: AdamW (lr=2e-5)

Learning Rate: Warmup + Decay

Regularization: Dropout (0.3)

Early Stopping: Enabled

Gradient Clipping: max_norm=1.0

Architecture Details

Encoder: Pre-trained XLM-RoBERTa

Pooling: [CLS] token representation

Classification Head: Single linear layer

Activation: None (raw logits)

Output: Probability distribution over 20 categories